

WHITEPAPER

Engineering Long-Life Battery-Powered Smart Water Meters

Part 5 — Battery Life Qualification and Validation

A Systems Engineering Approach to Battery Life in Advanced Metering Infrastructure

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Part	5 of 7 — Battery Life Qualification and Validation Whitepaper Series: ☐ Part 1 Understanding Battery Life and the Battery Life Budget ☐ Part 2 Battery Chemistry and Energy Storage ☐ Part 3 Ultra-Low-Power Hardware Architecture ☐ Part 4 Firmware as the Energy Manager ☒ Part 5 Battery Life Qualification and Validation ☐ Part 6 From Laboratory to Field: Achieving Real-World Battery Life ☐ Part 7 Practical Recommendations and the Future of Long-Life Smart Water Metering
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“Battery life is not demonstrated by calculation alone. It must be validated through disciplined engineering testing.”

5.1 Why Battery Qualification Matters

Battery life is one of the most important performance specifications of a smart water meter.

Utilities expect deployed meters to operate reliably for ten to fifteen years with minimal maintenance or battery replacement.

Because products cannot realistically be tested continuously for more than a decade before release, manufacturers must rely on engineering validation techniques to demonstrate that long-term performance targets are achievable.

Consequently, battery life should never be regarded as a theoretical calculation alone.

It should be supported by structured laboratory testing, environmental qualification, accelerated ageing, and field validation.

Only by combining these activities can engineers develop confidence that projected battery life will translate into real-world deployment performance.

Calculations Are Only the Starting Point

Battery life estimation begins with analysis.

Engineers calculate expected energy consumption based on:

- Sleep current
- Active current
- Communication profile
- Sensor operation
- Reporting frequency
- Expected battery capacity

These calculations form the initial Battery Life Budget.

However, every calculation depends on assumptions.

Examples include:

- Stable temperature
- Known communication conditions
- Typical battery characteristics
- Expected firmware behaviour

Real deployments rarely operate under ideal conditions.

Qualification testing is therefore required to validate whether those assumptions remain valid.

Engineering Validation Reduces Risk

Qualification testing provides confidence that the product will continue operating under realistic deployment conditions.

Rather than attempting to prove an exact battery lifetime, testing seeks to answer practical engineering questions such as:

- Does standby current remain within specification?
- Does communication behave correctly at low battery voltage?
- Does the product recover after temporary power interruptions?
- Does battery performance remain stable at high temperature?
- Do firmware updates complete successfully under realistic conditions?

These questions directly influence long-term reliability.

Qualification Covers the Entire System

Battery validation should not focus solely on the battery.

The complete system should be evaluated, including:

- Battery
- Power supply architecture
- Hardware
- Firmware
- Communication module
- Sensors
- PCB implementation
- Enclosure
- Environmental protection

Battery life is ultimately determined by the interaction of all these elements rather than by any single component.

Field Conditions Are Difficult to Reproduce

Laboratory testing provides repeatability.

Field deployment provides realism.

Both are necessary.

Laboratory testing offers:

- Controlled temperature
- Stable power measurement
- Repeatable communication scenarios
- Instrumentation access

Field deployment introduces:

- Variable network coverage
- Weather
- Underground installations
- Customer usage patterns
- Environmental ageing

A comprehensive qualification programme combines both approaches.

Validation Supports Utility Confidence

Utilities invest in AMI infrastructure expecting long operational lifetimes.

Battery qualification helps provide confidence that products have been evaluated under conditions representative of their intended deployment.

This reduces uncertainty during procurement and supports more informed engineering decisions throughout the project lifecycle.

Engineering Insight

Battery life is not a promise made at product launch. It is a hypothesis that must be continuously supported by engineering evidence.

Practitioner Observation

Many battery life discussions focus on theoretical current consumption calculations. While these calculations are essential during product design, experienced engineering teams recognise that laboratory measurements, environmental qualification, communication testing, and field observations are equally important. Long-life performance is established through the combination of analysis and validation rather than either approach alone.

Key Takeaways

- Battery calculations establish the initial Battery Life Budget.
- Qualification testing validates engineering assumptions.
- Battery validation should evaluate the complete AMI system.
- Laboratory and field testing provide complementary evidence.
- Long-term battery confidence is built through systematic engineering validation.

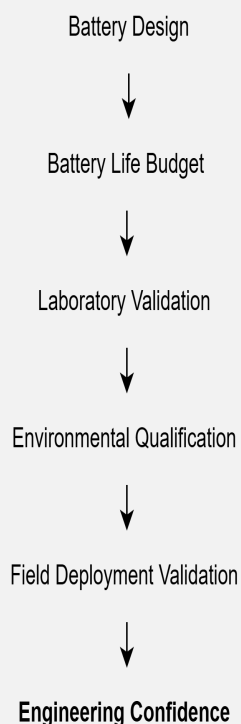


Figure 5.1 — Battery Qualification Methodology

Long-life battery performance is established through a structured qualification methodology that combines engineering analysis, laboratory validation, environmental testing, and real-world field experience.

5.2 Measuring Ultra-Low Current Accurately

Battery life calculations are only as accurate as the measurements that support them.

For smart water meters expected to operate for more than a decade, standby current is often measured in microamperes or even hundreds of nanoamperes.

At these current levels, seemingly insignificant measurement errors can lead to substantial inaccuracies in projected battery life.

Consequently, measuring ultra-low current requires specialised equipment, careful test methods, and disciplined engineering practice.

Small Measurement Errors Become Large Lifetime Errors

Consider a device designed to consume an average standby current of **2 μA** .

If the actual current is **3 μA** , the additional **1 μA** may appear insignificant during laboratory testing.

However, because this current flows continuously over many years, the cumulative energy loss becomes substantial.

An error of only a few microamperes can reduce battery life by many months or even years, depending on the application and battery capacity.

For long-life AMI products, accurate current measurement is therefore essential rather than optional.

Dynamic Current Cannot Be Measured with a Single Reading

Smart water meters rarely consume constant current.

Instead, current varies continuously depending on system activity.

Typical operating states include:

- Deep sleep
- RTC wake-up
- Sensor sampling
- Pulse counting
- Flash memory access
- Communication module startup
- Wireless transmission
- OTA firmware update

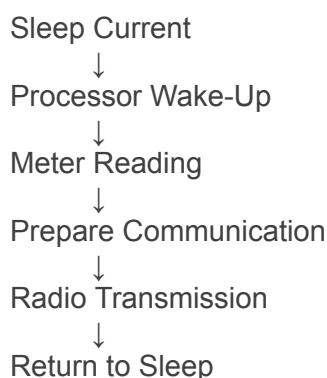
Each state exhibits different current characteristics.

Consequently, battery qualification should evaluate the complete current profile rather than relying solely on average current measurements.

Understand the Entire Current Profile

A typical operating cycle consists of several distinct phases.

For example:



Each phase contributes differently to total battery consumption.

Understanding the duration and current level of every phase enables engineers to identify where optimisation will have the greatest impact.

Appropriate Measurement Equipment

Different measurement tasks require different instruments.

Examples include:

Digital Multimeter (DMM)

Suitable for:

- Stable standby current
- Static current verification

Limitations:

- May not capture short-duration communication pulses.

Source Measure Unit (SMU)

Suitable for:

- Precision current measurement
- Controlled voltage supply
- Laboratory characterisation

Electronic Load

Useful for:

- Battery simulation
- Controlled discharge testing
- Validation under repeatable conditions

Oscilloscope with Current Probe

Suitable for:

- Dynamic current profiling
- Communication pulse analysis
- Wake-up timing investigation

No single instrument provides all the information required for comprehensive battery validation.

Minimise Measurement Disturbance

The measurement setup itself may influence the results.

Potential sources of error include:

- Lead resistance
- Instrument burden voltage
- Ground loops
- Probe capacitance
- Long wiring
- Poor shielding

Careful test fixture design helps ensure that the measurement system does not unintentionally alter device behaviour.

Measure Under Representative Conditions

Standby current measured on an engineering prototype inside a laboratory may differ from current measured on a production device installed inside its enclosure.

Validation should therefore consider:

- Production hardware
- Final firmware
- Actual battery
- Production PCB
- Operational enclosure
- Expected temperature range

These conditions provide results that are more representative of field deployment.

Repeatability Is More Important Than a Single Reading

Battery qualification should not rely on one measurement.

Instead, engineers should verify that results are:

- Repeatable
- Consistent across multiple samples
- Stable over time
- Reproducible using different test setups

Repeatability builds confidence that measured current genuinely represents product performance rather than measurement uncertainty.

Engineering Insight

In ultra-low-power engineering, measuring current accurately is often more challenging than reducing it. Confidence in battery life begins with confidence in the measurement itself.

Practitioner Observation

During product development, engineers sometimes optimize standby current based on a single laboratory measurement. However, repeat testing using different instruments, production hardware, or environmental conditions may reveal additional leakage paths or dynamic current behaviour that were not initially observed. Reliable battery qualification therefore depends on understanding both the measurement equipment and the device under test.

Key Takeaways

- Microamp-level measurement errors accumulate over many years.
- Smart water meters exhibit highly dynamic current profiles.
- Different instruments are required for static and dynamic measurements.
- Test fixtures should minimise measurement disturbance.
- Repeatable measurements provide greater confidence than isolated readings.

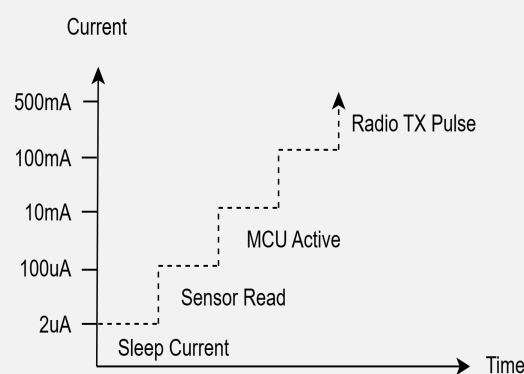


Figure 5.2 — Typical Current Profile of a Smart Water Meter

A smart water meter operates across multiple power states, from microamp-level sleep current to high-current communication bursts. Accurate battery qualification requires measurement of the complete current profile rather than a single average value.

5.3 Environmental Qualification: Temperature, Humidity and Real-World Conditions

A battery that performs well in a controlled laboratory environment may behave very differently after years of deployment in the field.

Smart water meters installed across ASEAN operate under environmental conditions that differ significantly from those assumed by many laboratory qualification tests.

High temperatures, prolonged humidity, seasonal flooding, underground installations, and continuous outdoor exposure all influence battery performance, electronic reliability, and overall system energy consumption.

Consequently, environmental qualification should evaluate the complete smart meter under realistic operating conditions rather than validating the battery in isolation.

Temperature Influences Every Part of the System

Temperature affects not only battery chemistry but also nearly every electronic component within the smart meter.

Examples include:

- Battery internal resistance
- Battery self-discharge
- Oscillator stability
- MCU timing accuracy
- RF performance
- Voltage regulator efficiency
- Sensor accuracy
- LCD performance (where applicable)

Battery life projections developed at a constant laboratory temperature should therefore be interpreted carefully when products will experience wide seasonal and daily temperature variations.

High Temperature Accelerates Battery Ageing

ASEAN deployments frequently expose smart water meters to elevated temperatures.

Examples include:

- Direct sunlight on exposed meter boxes
- Underground chambers with poor ventilation
- Roadside installations
- Concrete enclosures
- Metal meter cabinets

Higher temperatures generally accelerate:

- Battery self-discharge
- Chemical ageing
- Material degradation
- Seal deterioration

Even if the smart meter itself consumes the same electrical current, battery ageing may occur more rapidly under prolonged high-temperature exposure.

Battery life calculations should therefore incorporate realistic storage and operating temperature assumptions rather than relying solely on nominal laboratory specifications.

Low Temperature Creates Different Challenges

Although tropical climates dominate much of ASEAN, products may also encounter lower temperatures during transportation, storage, export, or deployment in cooler regions.

Low temperatures may result in:

- Increased battery internal resistance
- Reduced available capacity
- Lower pulse current capability
- Slower electrochemical reactions
- Temporary voltage depression during transmission

These effects are particularly relevant during high-current communication events such as NB-IoT transmission or OTA firmware updates.

Accordingly, qualification testing should evaluate both high- and low-temperature operating conditions where relevant to the intended deployment.

Humidity Is More Than a Mechanical Concern

Humidity is often associated with enclosure sealing and corrosion.

However, prolonged moisture exposure may also influence electrical performance.

Potential effects include:

- Surface leakage current
- Corrosion of exposed conductors
- Connector degradation
- Increased contact resistance
- Sensor contamination
- PCB insulation deterioration

Although these effects develop gradually, they may contribute to increased standby current or reduced communication reliability over many years of service.

Environmental qualification should therefore include long-duration humidity exposure representative of expected deployment conditions.

Flooding and Water Ingress

Many smart water meters are installed in underground chambers where flooding is a realistic operational risk.

Qualification should verify that the product continues to operate correctly after exposure to:

- Temporary immersion
- Splashing
- Heavy rainfall
- Standing water
- High-moisture environments

Testing should confirm not only enclosure integrity but also:

- Communication performance
- Battery compartment sealing
- Long-term insulation resistance
- Sensor reliability
- Recovery after drying

Ingress protection is valuable only if it can be maintained throughout the intended operational lifetime.

Environmental Cycling Is More Representative Than Constant Conditions

Real deployments rarely experience constant temperature or humidity.

Instead, products undergo repeated environmental cycles.

Examples include:

- Day-to-night temperature variation
- Seasonal weather changes
- Dry and wet periods
- Repeated condensation and evaporation

Repeated cycling may introduce stresses that are not apparent during steady-state laboratory testing.

Qualification programmes should therefore include temperature and humidity cycling wherever practical.

Test the Complete System

Environmental testing should evaluate the assembled smart meter rather than individual components alone.

The test article should represent the final production configuration, including:

- Battery
- PCB
- Firmware
- Enclosure
- Antenna
- Seals
- Connectors
- Meter body

Only complete-system testing can reveal interactions between environmental conditions and overall product performance.

Engineering Insight

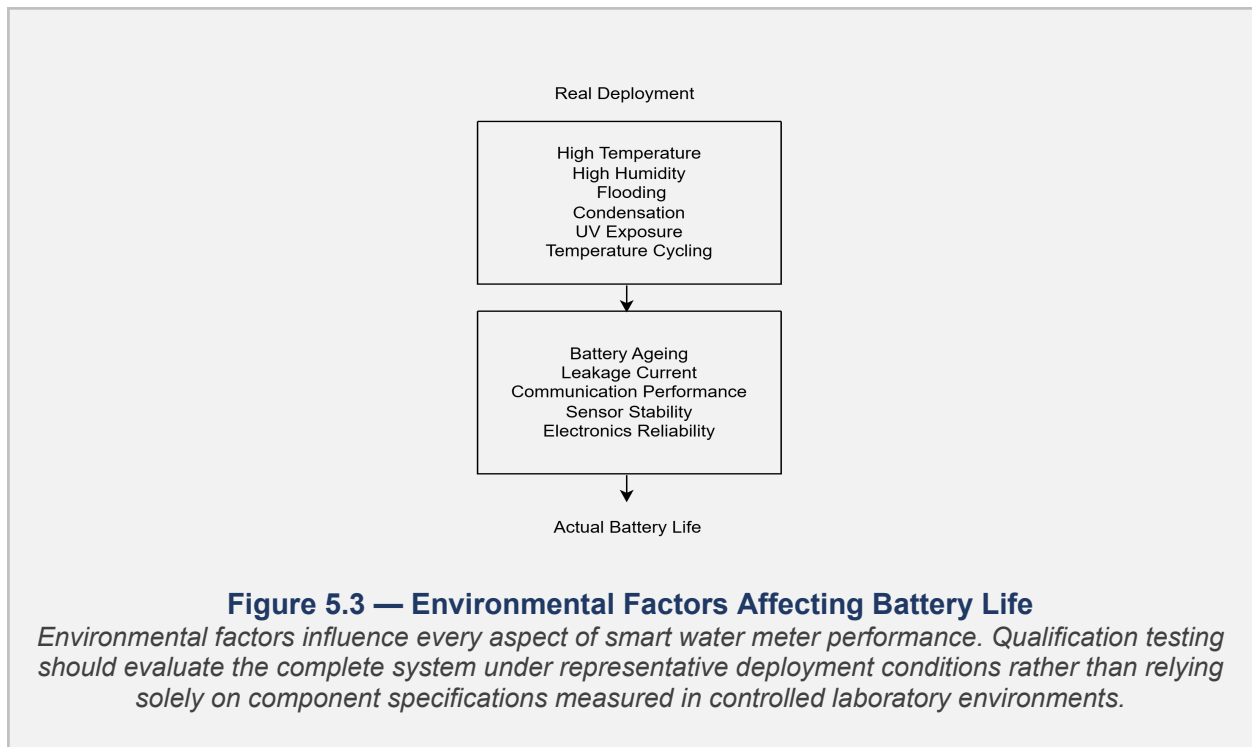
Environmental qualification validates the product that utilities deploy—not the components that suppliers specify. Long-life battery performance depends on how the complete system responds to years of real-world environmental exposure.

Practitioner Observation

Laboratory testing performed at room temperature often confirms that a design meets its electrical specifications. However, field deployments across tropical environments may reveal additional influences such as higher battery self-discharge, increased communication retries caused by condensation or enclosure effects, and gradual increases in leakage current due to humidity. Environmental qualification helps identify these behaviours before mass deployment rather than after thousands of meters have been installed.

Key Takeaways

- Temperature influences both battery chemistry and electronic performance.
- High temperatures accelerate battery ageing even when current consumption remains unchanged.
- Humidity may affect electrical leakage as well as mechanical reliability.
- Flood-prone installations require complete-system validation rather than enclosure testing alone.
- Environmental cycling provides more representative qualification than constant laboratory conditions.



5.4 Current Profiling and Communication Pulse Analysis

Battery life is ultimately determined by the energy consumed over time.

Although average current provides a convenient summary, it often hides the dynamic behaviour that dominates power consumption in modern smart water meters.

Wireless communication, processor wake-up, Flash memory programming, sensor operation, and OTA firmware updates all generate short-duration current pulses that may be hundreds or even thousands of times higher than standby current.

Understanding these transient events is essential for accurate battery qualification.

Current profiling therefore forms a critical part of validating the Battery Life Budget.

Average Current Does Not Tell the Whole Story

Battery life calculations frequently begin by estimating average current consumption.

While useful, average current alone cannot explain:

- Peak current demand
- Communication pulse duration
- Processor wake-up behaviour

- Battery voltage recovery
- Power supply stability
- Radio startup characteristics

Two products may exhibit the same average current while having completely different current profiles.

Their real-world battery performance may therefore differ significantly.

Current Changes Throughout Every Operating Cycle

A smart water meter continuously transitions between different operating states.

A typical reporting cycle may include:

1. Deep sleep
2. RTC wake-up
3. MCU initialisation
4. Meter measurement
5. Sensor processing
6. Data preparation
7. Communication module startup
8. Network attachment
9. Data transmission
10. Acknowledgement reception
11. Return to sleep

Each stage exhibits a unique current signature.

Profiling the complete operating cycle provides engineers with a far better understanding of where battery energy is actually consumed.

Communication Pulses Dominate Energy Consumption

For cellular AMI systems such as NB-IoT or LTE-M, the highest current demand generally occurs during radio transmission.

Typical communication events may involve:

- RF power amplifier activation
- Network synchronisation
- Authentication
- Packet transmission
- Reception of acknowledgements

Although these events may last only a short time, they often consume significantly more energy than hours of standby operation.

Optimising communication behaviour therefore offers one of the greatest opportunities for improving battery life.

Pulse Duration Matters as Much as Pulse Height

Peak current alone does not determine energy consumption.

A very high current pulse lasting only a few milliseconds may consume less energy than a lower current maintained for several seconds.

Engineers should therefore evaluate:

- Peak current
- Pulse duration
- Pulse frequency
- Recovery time
- Total energy per communication event

The Battery Life Budget is determined by the combination of these characteristics rather than by any single measurement.

Identify Opportunities for Optimisation

Current profiling helps engineers answer practical questions such as:

- Does the processor remain active longer than necessary?
- Is communication taking longer than expected?
- Are retries occurring?
- Are peripherals powered when not required?
- Is Flash programming efficient?
- Is the radio shutting down correctly?

Many battery improvements are discovered through current profile analysis rather than through schematic review alone.

Compare Firmware Versions

Current profiling is also valuable during firmware development.

Comparing profiles before and after firmware changes helps engineers determine whether new functionality has altered energy consumption.

Typical comparisons include:

- Before and after communication optimisation
- Before and after OTA implementation
- Before and after new alarm features
- Before and after protocol upgrades

Visual comparison often reveals unexpected increases in processor activity or communication duration that are not obvious from source code inspection.

Correlate Current Profiles With Battery Models

Current profiling should not be performed in isolation.

Measured current waveforms should be incorporated into battery modelling and Battery Life Budget calculations.

This allows engineers to evaluate:

- Daily energy consumption
- Expected annual battery usage
- Remaining battery margin
- Sensitivity to reporting frequency
- Sensitivity to communication retries

Combining measurement with modelling produces more realistic battery life predictions than either technique alone.

Engineering Insight

Batteries respond to real current pulses—not average current. Understanding the complete current profile reveals where battery energy is actually consumed and where meaningful optimisation is possible.

Practitioner Observation

In many development projects, battery optimisation begins by reducing standby current. However, detailed current profiling often reveals that communication events dominate overall energy consumption. Improvements that shorten radio activity, reduce retries, or minimise processor active time during communication frequently produce larger battery life gains than further reductions in already low sleep current.

Key Takeaways

- Average current should be supported by full current profile analysis.
- Communication pulses are often the largest energy events.
- Pulse duration is as important as peak current.
- Current profiling identifies optimisation opportunities that average measurements cannot reveal.
- Measured current profiles should be incorporated into Battery Life Budget calculations.

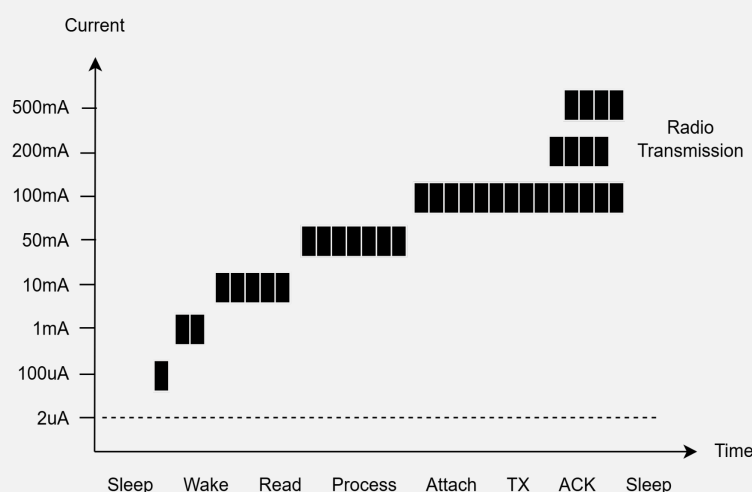


Figure 5.4 — Typical Smart Meter Current Profile

Current profiling reveals the complete operating cycle of a smart water meter, highlighting the relative contribution of sleep current, processor activity, network attachment, and wireless transmission to overall battery consumption.

5.5 Accelerated Battery Ageing: What It Can—and Cannot—Tell You

Smart water meters are typically designed to operate continuously for ten to fifteen years.

Waiting for an actual deployment to reach the end of its service life before validating battery performance is clearly impractical.

Manufacturers therefore rely on accelerated ageing techniques to evaluate long-term battery behaviour within a manageable engineering timeframe.

Accelerated testing provides valuable insight into battery degradation mechanisms, component reliability, and system robustness.

However, it is important to recognise that accelerated ageing is a predictive engineering tool rather than a direct substitute for long-term field operation.

Its purpose is to improve engineering confidence—not to recreate every aspect of ten years of real-world deployment.

Why Accelerated Testing Is Necessary

Long-life products must often be qualified before large-scale production begins.

Utilities, regulators, and manufacturers cannot wait a decade to determine whether a product is suitable for deployment.

Accelerated testing allows engineering teams to evaluate:

- Battery ageing trends
- Material degradation
- Environmental robustness
- Firmware stability over time
- Component reliability
- System performance under elevated stress

The objective is to expose potential weaknesses early enough for corrective action before mass deployment.

Temperature Is the Most Common Acceleration Method

Many battery ageing studies use elevated temperature to accelerate chemical reactions.

Higher temperatures generally increase:

- Chemical reaction rates
- Self-discharge
- Material ageing
- Electrolyte degradation
- Seal deterioration

This allows engineers to observe long-term degradation trends within a shorter test period.

However, elevated temperature does not perfectly replicate natural ageing under normal operating conditions.

It simply accelerates selected degradation mechanisms.

Not Every Failure Accelerates Equally

A common misconception is that all ageing processes respond similarly to higher temperatures.

In reality, different failure mechanisms behave differently.

Examples include:

Failure Mechanism	Temperature Acceleration
Battery self-discharge	High
Chemical ageing	High

Seal degradation	Moderate
PCB corrosion	Depends on humidity
RF communication issues	Little direct acceleration
Firmware defects	No acceleration

This illustrates why no single accelerated test can validate every aspect of long-term product performance.

A comprehensive qualification programme requires multiple complementary tests.

Accelerated Testing Cannot Simulate Every Field Condition

Laboratory ageing cannot fully reproduce:

- Variable network coverage
- Customer water usage
- Installation quality
- Underground flooding
- Mechanical damage
- Seasonal operating patterns
- Utility maintenance practices

These operational factors influence real battery life but are difficult to compress into accelerated laboratory testing.

Consequently, field validation remains an essential part of battery qualification.

Combine Multiple Qualification Methods

The most reliable battery validation programmes combine several approaches.

Typical examples include:

- Analytical Battery Life Budget calculations
- Laboratory current measurements
- Environmental qualification
- Accelerated ageing
- Communication stress testing
- Pilot field deployments
- Long-term production monitoring

Each method contributes different evidence.

Together, they provide a much stronger basis for engineering confidence than any individual test alone.

Monitor Degradation Trends Rather Than Absolute Lifetime

Accelerated testing is particularly useful for identifying trends.

Engineers may monitor:

- Capacity retention
- Internal resistance
- Pulse current capability
- Communication stability
- Standby current drift
- Firmware behaviour
- Mechanical integrity

Trend analysis often provides more useful engineering information than attempting to predict an exact service lifetime.

Engineering Judgement Remains Essential

Accelerated testing produces valuable data.

Interpreting that data requires engineering judgement.

Results should always be considered alongside:

- Battery chemistry
- Deployment environment
- Communication profile
- Firmware behaviour
- Operational assumptions
- Utility requirements

Battery life prediction is therefore an engineering assessment supported by evidence rather than a single mathematical calculation.

Engineering Insight

Accelerated ageing increases engineering confidence—it does not eliminate engineering uncertainty. Long-life battery qualification is built from multiple sources of evidence rather than a single laboratory test.

Practitioner Observation

Experienced engineering teams rarely rely on accelerated ageing alone when evaluating long-life battery performance. Instead, they compare accelerated ageing results with measured current profiles, Battery Life Budget calculations, environmental qualification

data, and pilot deployment observations. Agreement between these independent sources provides far greater confidence than any individual test result.

Key Takeaways

- Accelerated ageing helps evaluate long-term degradation within practical development timescales.
- Elevated temperature accelerates many—but not all—battery ageing mechanisms.
- No accelerated test can fully reproduce every real-world deployment condition.
- Trend analysis is often more valuable than attempting to predict an exact service lifetime.
- Reliable battery qualification combines analytical, laboratory, environmental, and field evidence.

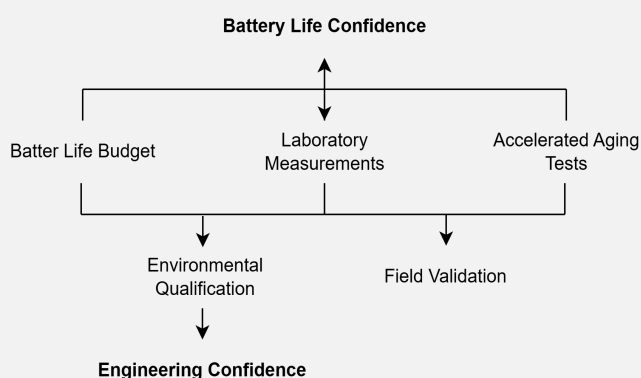


Figure 5.5 — Multiple Sources of Battery Life Validation

Long-life battery confidence is established by combining analytical modelling, laboratory measurements, accelerated ageing, environmental qualification, and field validation rather than relying on any single test method.

5.6 Laboratory Testing Versus Field Validation

Laboratory testing and field validation serve different but equally important purposes during the development of long-life smart water metering systems.

Laboratory testing provides controlled, repeatable conditions that allow engineers to evaluate individual aspects of product performance.

Field validation, by contrast, exposes the complete system to the operational uncertainties that cannot be fully reproduced inside a laboratory.

Neither approach is sufficient on its own.

Reliable battery qualification requires both controlled engineering verification and practical deployment experience.

The Strength of Laboratory Testing

Laboratory testing allows engineers to isolate variables and evaluate product behaviour under carefully controlled conditions.

Typical laboratory activities include:

- Standby current measurement
- Current profiling
- Battery characterisation
- Temperature testing
- Humidity testing
- Communication verification
- Firmware validation
- Environmental qualification

Because test conditions remain repeatable, engineers can accurately compare different hardware revisions, firmware versions, or battery configurations.

This repeatability makes laboratory testing indispensable during product development.

Laboratory Conditions Have Practical Limits

Despite its advantages, laboratory testing cannot fully reproduce real deployment environments.

For example, laboratories cannot realistically simulate:

- Changing cellular network performance
- Customer water usage patterns
- Installation workmanship
- Underground chamber conditions
- Seasonal weather variation
- Utility operational procedures
- Infrastructure ageing over many years

Consequently, laboratory testing should be viewed as controlled engineering verification rather than a complete representation of field operation.

Field Validation Reveals Operational Reality

Pilot deployments expose products to the environments in which they will actually operate.

During field validation, engineers can observe:

- Communication reliability
- Daily data availability
- Installation variability
- Environmental exposure
- Utility operating procedures
- Customer interaction
- Long-term battery behaviour
- Maintenance requirements

These observations often identify practical issues that were not apparent during laboratory qualification.

Field Testing Should Challenge the Design

A common mistake is selecting only favourable pilot locations.

Successful field validation should intentionally include challenging deployment environments such as:

- Underground meter chambers
- High-rise residential buildings
- Weak cellular coverage areas
- Flood-prone installations
- Dense urban environments
- Rural edge-of-network locations

Testing only ideal sites provides limited confidence for large-scale deployment.

A representative pilot should reflect the diversity of conditions expected during full rollout.

Compare Laboratory Predictions With Field Measurements

One objective of field validation is to confirm whether engineering assumptions remain valid under operational conditions.

Useful comparisons include:

Laboratory Measurement	Field Observation
Sleep current	Battery voltage trend
Communication energy	Actual upload success rate

Retry assumptions	Real retry frequency
RF performance	Measured signal quality
Battery model	Actual battery behaviour

Agreement between laboratory predictions and field observations increases confidence in the Battery Life Budget.

Where significant differences exist, engineers should investigate the underlying causes before mass deployment.

Field Data Should Improve Future Designs

Field validation is not simply a final acceptance activity.

It should also provide feedback for future product improvements.

Examples include:

- Optimising firmware scheduling
- Refining retry algorithms
- Improving antenna design
- Updating Battery Life Budget assumptions
- Enhancing installation procedures
- Improving communication strategies

This continuous learning process strengthens both current and future product generations.

Qualification Continues After Product Release

Even after commercial deployment, engineering teams should continue monitoring field performance.

Useful operational indicators include:

- Battery replacement rate
- Communication success rate
- Daily data availability
- Firmware update success
- Alarm performance
- Unexpected power consumption trends

Ongoing field monitoring provides early warning of emerging issues and helps validate long-term battery life assumptions throughout the product lifecycle.

Engineering Insight

Laboratory testing proves that a product works under controlled conditions. Field validation demonstrates that it continues to work under real operating conditions. Long-life battery confidence requires both.

Practitioner Observation

Engineering teams occasionally discover that products meeting all laboratory qualification requirements behave differently after deployment. Communication retries may occur more frequently than anticipated, installation practices may introduce additional RF attenuation, or local environmental conditions may influence battery performance. Well-designed pilot programmes help identify these factors before full-scale deployment, allowing design assumptions to be refined using real operational evidence.

Key Takeaways

- Laboratory testing provides repeatable engineering validation.
- Field validation confirms performance under real operating conditions.
- Pilot deployments should intentionally include challenging installation environments.
- Comparing laboratory predictions with field measurements strengthens confidence in battery life estimates.
- Continuous field monitoring supports ongoing improvement throughout the product lifecycle.

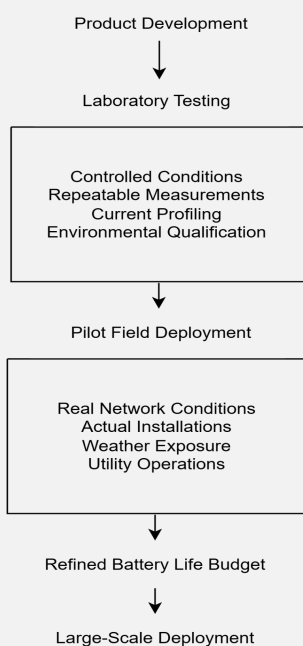


Figure 5.6 — Laboratory Qualification and Field Validation

Laboratory qualification and field validation complement one another. Controlled testing establishes engineering performance, while pilot deployments verify that those results remain valid under real operating conditions before large-scale rollout.

5.7 Production Testing and Manufacturing Consistency

Battery qualification is often performed using engineering prototypes during product development.

However, long-term product performance ultimately depends on the consistency of every production unit delivered to the customer.

Even a well-designed smart water meter may fail to achieve its intended battery life if manufacturing variation introduces higher standby current, inconsistent communication performance, or assembly defects.

Consequently, battery life qualification should extend beyond prototype validation to include manufacturing quality assurance throughout the product lifecycle.

Prototype Success Does Not Guarantee Production Success

Engineering prototypes are typically assembled under carefully controlled conditions.

Production introduces additional sources of variation, including:

- Component manufacturing tolerances
- PCB assembly quality
- Soldering consistency
- Firmware programming
- Battery installation
- Mechanical assembly
- Calibration procedures
- Production test coverage

Each variation may influence battery performance, even when individual components remain within specification.

Maintaining consistent manufacturing processes is therefore essential for achieving predictable long-term battery life.

Manufacturing Variation Can Affect Battery Performance

Small production differences may alter system energy consumption in subtle ways.

Examples include:

- Higher leakage current due to contamination
- Poor solder joints increasing electrical resistance
- Incorrect firmware configuration
- Misaligned RF antennas reducing communication efficiency
- Improper battery installation
- Defective sealing allowing moisture ingress

While these issues may not cause immediate product failure, they can gradually reduce operational lifetime.

Production quality should therefore be regarded as part of battery engineering rather than solely manufacturing control.

Production Testing Should Verify Low-Power Performance

Manufacturing tests frequently focus on basic functionality.

Typical checks include:

- Meter operation
- Communication
- Display operation
- Calibration
- Sensor functionality

For long-life battery products, production testing should also verify critical low-power characteristics where practical.

Examples include:

- Sleep current within specification
- Wake-up functionality
- RTC operation
- Battery voltage measurement
- Communication startup
- Power supply stability

Detecting abnormal power consumption before shipment helps prevent long-term field issues.

Firmware Configuration Must Remain Controlled

Firmware configuration has a direct influence on battery life.

Production processes should ensure that every device receives:

- Correct firmware version
- Correct communication parameters
- Appropriate reporting interval
- Approved retry configuration
- Security credentials
- Production calibration data

Configuration errors introduced during manufacturing may increase battery consumption even when the hardware is functioning correctly.

Strict version control and automated programming procedures help minimize these risks.

Statistical Process Control Supports Consistency

Rather than evaluating only individual units, manufacturers should monitor production trends.

Examples include:

- Average standby current
- Current distribution
- Communication success rate
- RF output verification
- Battery voltage consistency
- Functional test yield

Monitoring statistical trends enables manufacturing teams to identify gradual process changes before they affect large production volumes.

Manufacturing Feedback Improves Future Designs

Production testing should not be viewed only as a quality gate.

It should also provide engineering feedback.

Recurring production issues may indicate opportunities to improve:

- PCB layout
- Component selection
- Assembly processes
- Test coverage
- Firmware robustness
- Mechanical design

A strong feedback loop between manufacturing and engineering contributes to continuous improvement across future product generations.

Engineering Insight

A battery life claim applies to every meter shipped—not just the engineering prototype. Manufacturing consistency is therefore an essential part of long-life battery performance.

Practitioner Observation

Engineering teams often invest significant effort optimising prototype power consumption, yet long-term field performance ultimately depends on production consistency. Even small increases in standby current or communication retries across thousands of manufactured units can have a measurable impact on lifecycle performance. Incorporating low-power verification into manufacturing quality processes helps ensure that production devices continue to meet the Battery Life Budget established during product development.

Key Takeaways

- Prototype qualification should be followed by robust production quality assurance.
- Manufacturing variation can influence long-term battery performance.
- Production testing should include verification of critical low-power functions.
- Firmware configuration control is essential for consistent battery life.
- Statistical process monitoring supports long-term manufacturing consistency.

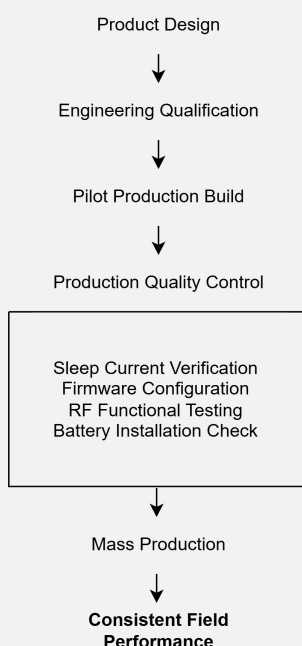


Figure 5.7 — Battery Quality Throughout the Manufacturing Lifecycle

Long-term battery performance depends on maintaining consistency from engineering qualification through mass production. Manufacturing quality assurance helps ensure that every deployed meter meets the intended Battery Life Budget.

5.8 Common Battery Qualification Mistakes

Battery qualification is intended to reduce uncertainty before large-scale deployment.

However, qualification programmes sometimes provide a false sense of confidence because important assumptions remain untested or unrealistic.

Many long-term battery problems observed in field deployments originate not from battery technology itself, but from incomplete validation methodologies.

Understanding these common qualification mistakes helps manufacturers design more effective validation programmes and enables utilities to evaluate battery life claims more critically during procurement.

Mistake 1 — Testing Only at Room Temperature

One of the most common qualification errors is validating battery performance exclusively under laboratory conditions of approximately 25°C.

While room-temperature testing provides a useful engineering baseline, it does not represent the environments encountered during actual deployment.

Smart water meters may experience:

- Tropical daytime temperatures
- Underground chambers
- Seasonal weather variations
- Flood-prone installations
- Cold storage during transportation
- Direct solar heating

Battery qualification should therefore evaluate representative environmental conditions rather than relying solely on room-temperature measurements.

Mistake 2 — Measuring Average Current Only

Some qualification programmes focus exclusively on average current consumption.

This approach overlooks important transient events such as:

- Processor wake-up
- Communication startup
- NB-IoT network attachment
- Radio transmission
- Flash programming
- OTA firmware updates

Battery energy is consumed during these dynamic events.

Comprehensive current profiling provides a far more accurate understanding of real energy consumption than average current measurements alone.

Mistake 3 — Ignoring Communication Conditions

Battery life is closely linked to communication performance.

Testing under excellent RF conditions may underestimate battery consumption in actual deployments where devices experience:

- Weak cellular coverage
- Underground installations
- High-rise buildings
- RF attenuation
- Network congestion
- Communication retries

Qualification programmes should therefore evaluate communication performance across a range of realistic signal conditions.

Mistake 4 — Assuming Battery Specifications Guarantee System Performance

Battery datasheets provide valuable technical information.

However, battery life depends on the complete smart meter system rather than the battery alone.

Examples include:

- Hardware design
- Firmware efficiency
- Communication behaviour
- Temperature
- Installation environment
- Battery management strategy

A high-quality battery cannot compensate for inefficient system design.

Battery qualification should therefore evaluate complete product performance.

Mistake 5 — Using Ideal Communication Profiles

Some battery calculations assume:

- Every transmission succeeds
- No retries occur
- Stable network coverage
- Constant signal strength

Real deployments rarely behave this way.

Battery qualification should include realistic assumptions regarding:

- Retry frequency
- Variable signal quality
- Communication delays
- Network recovery

Including these operational factors produces more representative Battery Life Budget calculations.

Mistake 6 — Treating Qualification as a One-Time Activity

Battery validation sometimes ends once engineering qualification is complete.

However, products continue evolving through:

- Firmware updates
- Production changes
- Component substitutions

- Communication network evolution
- Utility operational changes

Qualification should therefore continue throughout the product lifecycle rather than ending at product release.

Mistake 7 — Confusing Marketing Claims With Engineering Evidence

Battery life is frequently presented as a single headline specification.

For example:

"15-year battery life."

Without supporting assumptions, this statement provides little engineering value.

Meaningful battery life claims should clearly define:

- Reporting interval
- Communication technology
- RF conditions
- Operating temperature
- Battery chemistry
- Data availability target
- Expected retry behaviour

Transparent assumptions allow meaningful comparison between different products and technologies.

Build a Qualification Programme Around Evidence

An effective qualification programme combines multiple independent sources of engineering evidence.

These include:

- Battery Life Budget calculations
- Current profiling
- Environmental qualification
- Accelerated ageing
- Communication testing
- Pilot field validation
- Production quality assurance

Agreement across these activities provides substantially greater confidence than any single test alone.

Engineering Insight

The objective of qualification is not to prove that a battery will last fifteen years. It is to demonstrate that the engineering assumptions supporting that expectation have been systematically verified.

Practitioner Observation

Battery qualification programmes are most effective when they challenge engineering assumptions rather than confirm them. Testing under difficult RF conditions, elevated temperatures, realistic reporting schedules, and production hardware often reveals opportunities for improvement that remain hidden during ideal laboratory testing. Engineers should therefore view qualification as a process of discovering weaknesses early, when they are still inexpensive to correct.

Key Takeaways

- Qualification should evaluate representative deployment conditions rather than ideal laboratory scenarios.
- Dynamic current profiling provides more insight than average current alone.
- Communication performance is a major contributor to battery life.
- Battery qualification should assess the complete smart meter system.
- Credible battery life claims require transparent engineering assumptions.

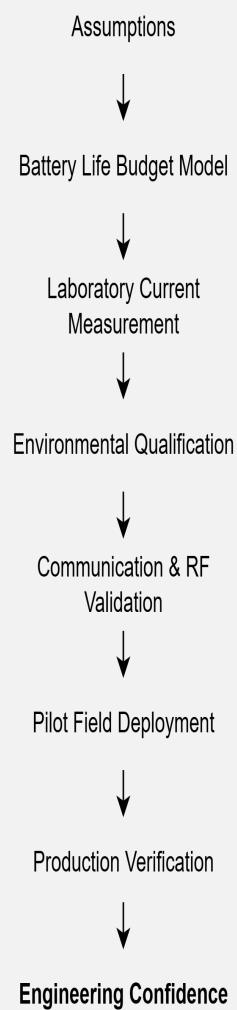


Figure 5.8 — From Assumptions to Engineering Evidence

Reliable battery life predictions are established by progressively validating engineering assumptions through laboratory testing, environmental qualification, communication analysis, field deployment, and production verification.

Summary

Designing a smart water meter for a ten- to fifteen-year operational life requires far more than selecting a high-capacity battery.

Long-term battery performance must be demonstrated through disciplined engineering validation that combines analytical modelling, laboratory measurement, environmental qualification, production quality assurance, and field experience.

Throughout this chapter, we have shown that battery qualification extends well beyond simple current measurements. Accurate validation requires understanding dynamic current profiles, evaluating communication behaviour, assessing environmental influences, monitoring production consistency, and recognising both the capabilities and limitations of accelerated ageing techniques.

Importantly, no single laboratory test can prove that a battery will achieve its intended service life. Instead, confidence is built through multiple independent sources of engineering evidence that consistently support the assumptions contained within the Battery Life Budget.

The most successful engineering organisations treat battery qualification as a continuous process rather than a one-time development milestone. As firmware evolves, communication networks change, manufacturing processes mature, and field experience accumulates, battery life assumptions should be reviewed and refined accordingly.

For utilities, understanding how battery qualification is performed is equally important. Procurement decisions should be based not only on advertised battery life but also on the quality of the engineering evidence supporting those claims. Transparent assumptions, representative testing, and measurable operational data provide a much stronger foundation for long-term project success than headline specifications alone.

Ultimately, battery qualification is not about proving perfection.

It is about reducing uncertainty through systematic engineering.

That disciplined approach enables manufacturers to design more reliable products, utilities to make more informed procurement decisions, and AMI deployments to achieve the long operational lifetimes expected from modern smart water infrastructure.

Key Messages

The principal lessons from this chapter include:

- Battery life predictions must be supported by structured engineering validation.
- Accurate current measurement requires appropriate instrumentation and repeatable test methods.
- Environmental qualification should reflect realistic deployment conditions rather than ideal laboratory environments.
- Current profiling provides deeper insight than average current measurements alone.
- Accelerated ageing improves engineering confidence but does not replace long-term field validation.
- Pilot deployments are essential for confirming laboratory assumptions under operational conditions.

- Manufacturing consistency is critical to ensuring that every production unit meets the intended Battery Life Budget.
- Credible battery life claims require transparent assumptions and supporting engineering evidence.

Engineering Confidence



Field Validation



Production Verification



Environmental Qualification



Laboratory Measurements



Battery Life Budget Model



Engineering Assumptions

Figure 5.9 — The Battery Qualification Pyramid

Confidence in long-life battery performance is built progressively. Engineering assumptions are strengthened through modelling, laboratory validation, environmental testing, manufacturing verification, and real-world field deployment.

Disclaimer

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About the Author



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Vincent Wong brings over **25 years of engineering and product leadership experience** across industrial infrastructure and technology sectors, with the past decade focused on smart metering, Advanced Metering Infrastructure (AMI), and NB-IoT deployment strategies in ASEAN utility environments. As a senior engineering and product development executive, his work spans hardware engineering, connectivity systems, and utility digitalization initiatives. His writing focuses on the practical operational, connectivity, deployment, and governance realities of smart utility programs in Southeast Asia.

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